Debdip Bandyopadhyay

Senior SDET | Enterprise Test Automation & AI Quality Engineering Lead

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PROFESSIONAL SUMMARY

Senior Software Development Engineer in Test (SDET) with **11+ years of experience** architecting high-scale enterprise test automation frameworks, microservice API validation suites, and AI quality engineering ecosystems across **Global Enterprise Systems, Generative AI Platforms, and Safety-Critical Aerospace (The Boeing Company)**. Deep technical mastery in **Python, Core Java, Playwright, Selenium, REST API testing (Requests, RestAssured), and Dockerized parallel execution grids** slashing regression cycles by ~**40%**. Core member of Cognizant's GenAI CoE leading automated testing for AI/ML workflows, LLM-as-a-Judge evaluation, and CI/CD quality gates. Proven expertise in **first-level root-cause log triage, performance/load testing (JMeter, Locust), and multi-functional team leadership**. Master's degree from **IIT Jodhpur (School of AI & Data Science)** with independent published research in foundation model evaluation (Elsevier SSRN DOIs: 10.2139/ssrn.7374319, 10.2139/ssrn.7374338, 10.2139/ssrn.7374378 & CERN Zenodo DOIs: 10.5281/zenodo.22177214, 10.5281/zenodo.22113940).

TECHNICAL SKILLS

- **Automation Frameworks & Languages:** Python (Expert), Core Java, Playwright, Selenium WebDriver, Pytest, Pytest-BDD, RestAssured, Karate, C++, Page Object Model (POM), Parallel Execution (pytest-xdist).
- **API, Microservices & Performance:** RESTful Web Services Testing, Requests, Postman (Path v12), Contract Testing (OpenAPI), Performance & Load Testing (JMeter, Locust, Gatling), Resiliency Testing, Microservices Architecture.
- **DevOps, CI/CD & Cloud:** Azure DevOps Pipelines, Docker, Docker Compose (Parallel Testbeds), Git, Maven, Linux / Bash, Allure Merged Reporting, Zero-Escape Quality Gates.
- **AI Quality & Enterprise Systems:** AI/ML Applications Testing, DeepEval, LLM-as-a-Judge, RAG Validation, Semantic Faithfulness Scoring, Root Cause Log Analysis, Safety-Critical Verification (Aerospace / DO-178C standards).

OPEN-SOURCE AI SYSTEMS & GITHUB PROJECTS

NVIDIA NIM-Powered LLM Verification Framework — github.com/debdipARVR/chatbot_tests_deepeval_bitnet

[Custom DeepEval LLM-as-a-Judge evaluator backed by NVIDIA NIM API (integrate.api.nvidia.com)]

- Engineered custom **NvidiaLLM** evaluator integrating NVIDIA NIM microservices with DeepEval's G-Eval framework.
- Automated semantic verification, factual correctness matching, and hallucination scoring on non-deterministic LLM responses.

Distributed Dockerized Parallel Test Grid — github.com/debdipARVR/debdipARVR-Testing

[Multi-container parallel test architecture with dynamic execution sharding and merged Allure reporting]

- Architected containerized parallel test runners orchestrating multi-container execution nodes via Docker Compose.
- Implemented automated test artifact consolidation and multi-container report aggregation for high-scale regression suites.

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions — **Apr 2021 — Present**

Lead SDET / Senior Test Automation Engineer — GenAI CoE Team — *Kolkata, India*

- **Enterprise QE & GenAI CoE Leadership:** Core SDET lead in the GenAI Center of Excellence, architecting test automation frameworks and evaluation ecosystems for customer-facing web applications, microservices, and AI-enabled workflows.
- **Playwright & Python Framework Architecture:** Designed and maintained enterprise-grade Playwright + Python + Pytest-BDD automation frameworks adopting modular design patterns, running parallelized across multi-browser configurations.
- **API & Microservice Validation:** Engineered automated integration test suites for RESTful web services and backend microservices, validating response payloads, error codes, authentication schemes, and API contract compliance.
- **Performance & Parallel Execution:** Built Dockerized parallel test execution grids using pytest-xdist, cutting regression execution cycle times by ~**40%** across high-consequence enterprise suites.
- **CI/CD Quality Gates & Root-Cause Triage:** Integrated automated test suites into Azure DevOps CI/CD pipelines, automating build certification, log scanning for first-level root cause analysis, and failure triage.
- **Multi-Functional Collaboration & Mentorship:** Partnered with developers, product managers, and QA teams to define testing charters and acceptance criteria; mentored 10+ junior and mid-level test engineers on QE best practices.

The Boeing Company — **Oct 2018 — Apr 2021**

Software Engineer — Test Automation — *Bengaluru, India*

- **Safety-Critical Systems Verification:** Engineered automated verification suites for mission-critical aerospace avionics operating under strict zero-tolerance engineering standards.
- **High-Volume Regression Suites:** Maintained, profiled, and executed high-consequence regression test suites of 2,000+ automated test cases across multi-phase release cycles, eliminating release escapes.
- **Mentorship & Code Quality Checking:** Conducted architectural code reviews, mentored junior automation engineers on test hygiene, and enforced automated code quality checking.
- **Recognition:** Awarded **Boeing Pride Award** and **Star Award** (2019) for exceptional verification delivery and defect prevention under compressed schedules.

PROFESSIONAL EXPERIENCE (CONTINUED)

Accenture LLP **Oct 2014 — Oct 2018**
Test Automation Engineer *Kolkata / Noida, India*

- **Multi-Platform Automation:** Developed modular, data-driven automation frameworks across Web, API, and Desktop applications, automating Integration, Functional, and System Regression suites.
- **Tooling Engineering & Cost Optimization:** Designed and deployed an in-house One Click Capture test automation utility, eliminating manual capture overhead and reducing third-party tooling expenses by ~USD 8,000 annually.
- **Requirement Decomposition:** Collaborated with system architects and domain leads to translate complex product requirements into robust, automated test plans and scenarios.

PUBLICATIONS & AI RESEARCH PREPRINTS

Multi-Scale 4-Pass Sentence Cloze Infilling with Pass-Adaptive Dynamic Gating for Robust AI Text Detection
[Zero-shot black-box framework achieving 99.58% AUROC on RAID (ACL 2024), 99.98% on HC3, and 0.83% FPR on formal human academic prose across a 240-essay dual corpus via Kuhn-Munkres S_3 bipartite matching and continuous sigmoidal gating]
Elsevier SSRN: ssrn.com/abstract=7374319 | **SSRN DOI:** 10.2139/ssrn.7374319 | **CERN Zenodo DOI:** 10.5281/zenodo.22177214 | **Benchmarks:** github.com/debdipARVR/cloze_congruence_reproducibility_benchmark_kit

ClozeCongruence: Model Fingerprinting and Traceback Attribution in LLMs via Reverse Cloze Resonance
[Foundation model provenance traceback achieving 100.00% Top-1 attribution accuracy (Macro-F_1=1.0000) across 6 frontier LLMs including NVIDIA Nemotron-3 120B, GPT-4o, Claude 3.5, and Gemini 3.7 with +82.21% self-model resonance gap and 12D micro-stylometrics]
Elsevier SSRN: ssrn.com/abstract=7374338 | **SSRN DOI:** 10.2139/ssrn.7374338 | **CERN Zenodo DOI:** 10.5281/zenodo.22113940 | **Interactive Demo:** huggingface.co/spaces/DebdipCS/cloze-congruence-ai-forensics-hub-v1

The Cloze Scaling Laws: Prober Parameter Dynamics, Native Blank-Infilling Architecture, and Traceback
[Systematic empirical scaling study establishing monotonic prober recovery (12B to 120B) and GLM-5.2 native blank-infilling inductive advantage (+51.16% separation gap, Cohen's d=5.62, p=2.87e-15) across international SOTA benchmarks]
Elsevier SSRN: ssrn.com/abstract=7374378 | **SSRN DOI:** 10.2139/ssrn.7374378 | **CERN Zenodo DOI:** 10.5281/zenodo.22158286 | **Author Portfolio:** ssrn.com/author=12882248

EDUCATION

Master of Technology (M.Tech) in AR & VR **Jul 2022 — Dec 2025**
Indian Institute of Technology Jodhpur (School of Artificial Intelligence & Data Science) *Jodhpur, India*

- **Key Coursework:** Deep Learning, Generative AI, Natural Language Processing (NLP), MLOps & DLOps, C++, Data Structures & Algorithms, Big Data & Cloud Computing.
- **High-Performance Compute:** Hands-on model training, distributed simulation, and pipeline validation on high-performance compute clusters.

Bachelor of Technology (B.Tech) in Computer Science & Engineering **Apr 2010 — Jun 2014**
Greater Noida Institute of Technology (AKTU University) *Greater Noida, India*

Higher Secondary (Class XII) — CBSE [PCMB] **2010**
Kendriya Vidyalaya No. 1, Salt Lake (KV No. 1 SLK) *Kolkata, India*

CERTIFICATIONS

- **Postman API Testing Path (v12)**
- **ASDA-MIT Certification**
- **Google Crash Course on Python (Coursera)**